Zaber Ibn Abdul Hakim

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Education

- **Master's of Science in Computer Science** **Aug 2024 — May 2026**
  Virginia Polytechnic Institute & State University | *CGPA* : 4.00 (out of 4.00)

Experience

Capital One **Jun 2026 — Present**
Senior Data Scientist
- Developing a generalized evaluation framework to benchmark LLM and agentic system performance.
- Improve validation components for a production multi-agent system, verifying factual accuracy of generated outputs in high-volume customer-service workflows.

Virginia Polytechnic Institute & State University **Aug 2024 — Jun 2026**
Graduate Researcher
- Creating a large-scale multimodal (text-video-audio) evaluation dataset & LLM-based method for **event/entity extraction** and **relation extraction** (event matching/search) to reduce the information gap between modalities.
- Improving the object and part detection of LLMs using LLM post-training, i.e., **reinforcement learning (PPO, GRPO)**.
- Uncovering **security vulnerabilities in multimodal LLMs** via novel **adversarial attack** in realistic black-box setting.

Comcast **Jun 2025 — Aug 2025**
AI Research Intern
- Built distributed data pipeline using **PySpark** to process large-scale conversation datasets and added **LLM supervision** into **clustering** frameworks using **LangChain**, **LLM Tool/Function Calling** to improve the categorization.
- Designed a hierarchical, multi-layered clustering framework by integrating LLM-based verification and summarization into agglomerative clustering to **create descriptive cluster hierarchies**.
- Developed **LLM-based evaluation frameworks** for comparing the different unsupervised clustering methods.

IQVIA **Jun 2023 — Jul 2024**
Machine Learning Engineer
- Created IQVIA's AI-Assistant by **fine-tuning (SFT) LLaMA** model on business-specific use cases to enhance user experience and product engagement, which secured a 4.2 million USD initial contract.
- Finetuned LLM via **QLoRA** and **quantized with AutoAWQ** that reduced response time by 54% and memory usage by 73%.
- Designed and deployed production **LLM serving infrastructure** using vLLM on **AWS cloud**, implementing **autoscaling, monitoring, and GPU inference optimization**.
- Experimented **retrieval augmented generation (RAG)** based solution to accommodate fast changing information.
- Enhanced IQVIA's **recommendation algorithms** suite, generating over $10M in annual revenue, and contributed to the **deployment & MLOps pipeline** of these models using **Kubeflow & Kubernetes**.

Research Publications

- ***ZIA Hakim**, NH Sarker, AM Ishmam, C Thomas*, "M^{3}EC: A Multimodal Multidocument Benchmark for Event Extraction and Coreference Resolution," Empirical Methods in Natural Language Processing **(EMNLP 2026), Accepted**
- ***ZIA Hakim***, MA Rahman*, NH Sarker*, B Paul*, SA Fattah*, "SONICS: Synthetic Or Not–Identifying Counterfeit Songs," International Conference on Learning Representations (**ICLR 2025**), **Accepted**.
- *NH Sarker, **ZIA Hakim**, A Asgarov, CW Tang, A Ishmam, C Thomas*, "Immunizing Models Against Harmful Long-Horizon Fine-Tuning via Contractive Optimization Dynamics," Computer Vision and Pattern Recognition **(CVPR 2026), Highlight**.
- *AS Hussain, G Srivastava, AM Ishmam, **ZIA Hakim**, C Thomas*, "SoundBreak: A Systematic Study of Audio-Only Adversarial Attacks on Trimodal Models," Proceedings of Annual Meeting of the Association for Computational Linguistics **(ACL 2026), Accepted**.
- *AM Ishmam, NH Sarker, **ZIA Hakim**, C Thomas*, "LAMP: Learning Universal Adversarial Perturbations for Multi-Image Tasks via Pre-trained Models," Association for the Advancement of Artificial Intelligence (**AAAI 2025**), **Accepted**.
- *A Asgarov, K Narasimhan, NH Sarker, H Alomari, CW Tang, A Sivakumar, **ZIA Hakim**, S Mallampati, C Thomas*, "NEST: Narrative Event Structures in Time for Long Video Understanding," Empirical Methods in Natural Language Processing (**EMNLP 2026**), **Accepted**.
- *KS Mehrab, H Alomari, NH Sarker, CW Tang, **ZIA Hakim**, A Karpatne, C Thomas*, "Reasoning-Guided Part-Level Visual Grounding via Reinforcement Learning," European Conference on Computer Vision (**ECCV 2026**), **Accepted**.

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Skills

Languages & Frameworks	Python, C++, CUDA, Java, SQL, PyTorch, TensorFlow, Hugging Face, LangChain, FastAPI, ReactJS
GenAI Tooling	LLM Fine-tuning (SFT & QLoRA), RAG, LLM Alignment, Reinforcement Learning, Model Evaluation & Steering
DevOps & MLOps	Docker, Kubernetes, Kubeflow, vLLM CI/CD, AWS (S3, EC2, SageMaker, Lambda), Weights & Biases
Research Domains	Multimodal Information Extraction, Video & Audio Understanding, LLM Security & Robustness, Adversarial Attacks, Information Retrieval, Multimodal Grounding, Reinforcement Learning
Software Practices	Git, PyTest, GPU Programming, CUDA Programming, Microservice, Unit & Integration Testing, Agile Development

Awards

- **Best Student Team, 4th Place, Kaggle Gold Medal** , SIIM-FISABIO-RSNA COVID-19 Detection — Aug 2021
- **Joint Winner** , Kaggle Days x Z by HP World Championship 2022, Shanghai — Sept 2022
- **Winner** , Deep Chimpact — Nov 2021
- **Winner** , IEEE Signal Processing Cup — May 2022
- **First Runner-up** , IEEE Video & Image Processing Cup — Sept 2022